

cannot charge your mobile phone with your TV power supply or vice versa.

EcoSmart technology includes an extra circuitry on the chip that senses when a power supply is in a low-power (no-load or standby) state and provides controlled output with a mechanism called 'On-Off' rather than pulse-width modulation (PWM) used in switching power supplies.

PWM works perfectly for large loads and more of constant-load situations. But when you start reducing the load, the pulse width becomes smaller and smaller. This leads to two types of losses—switching loss and induction loss. At level '0' (no load), the efficiency drops significantly as the power supply design is optimised for full-load (maximum load) condition.

The On-Off technology is designed such that for every cycle, true power is delivered. If you don't need power, the cycle is eliminated. So it intelligently monitors the output voltage and controls the on-off for complete cycles. That means for no load, there will be no switching, hence no losses.

#### **Which applications are driving the growth of the power supply market and R&D?**

As one of the most important trends is energy efficiency, all the key applications which can be considered as the growth drivers are in and around this trend. The OEMs which are making products are using energy efficiency as a marketing tool.

One of the key applications is smart metering. It allows you to utilise the transmission lines efficiently. It provides detailed information about the energy usage and accordingly takes control action. You can read the meter remotely and also implement connectivity. You can also charge and penalise for theft of electricity and tampering of the meters. Designing power supply for these smart meters is also very critical as you install a meter and it consumes power forever. You don't want the meter to fail and it typically runs for 50 years.

The second biggest market is lighting. Until now lighting did not require power supply, but LED does. For a cellphone, who cares how long will the charger really work? People will throw it away and buy a new one. But in lighting and LEDs, quality and reliability are very important. As LEDs are claimed to have a life of ten years, you also want the power supply to work for that long or more.

#### **Why do you need a specific type of power supply for applications like LED lighting? How are these different from regular power supplies?**

LED drivers which you normally use are DC-DC power converters that operate at a low voltage and are suitable for applications where low-voltage DC is present, such as the back-light of the laptop. Another type of LED driver is AC-DC that gets plugged into the wall and is different from the above-mentioned. These are power supplies for LED ap-

plications. As they operate at a very low wattage (around 2 watts), they have different design considerations.

LED is the only application where you need to do the power factor correction (PFC) all the way down to low watts. In all other applications, the PFC does not start until 70 watts or so. Earlier, PFC wasn't considered important because the electronic load was very less compared to the total load on the grid. But in the last few years, the electronics load has grown so dramatically that now it's affecting utility companies.

For LED applications, you need to have independent PFC because when you convert AC into DC you have some input capacitance to smoothen the line. This is a reactive load which is inefficient. In other reactive loads, you balance it by using inductance or capacitance, but for electronics loads with switching power supply, you can't just put an inductor to do the power factor correction. You can do it much more elegantly by using an electronic circuit. This is what is included in most of the chips.

Another interesting challenge in LEDs is that these last so long that they outlast the power supply. LEDs can last about 15-20 years, while the power supply can last ten years. The power supply is made out of many components like electrolytic capacitors which have the shortest life time, and life of the optocoupler that provides isolation between the primary and secondary is also short.

#### **What is your take on analogue design in India?**

Analogue designers or hardware engineers are very difficult to get in India. Software engineers are the easiest to find. Next come digital design engineers and then the analogue engineers. In the US and other countries, there is a balance in the availability of hardware and software engineers.

One of the reasons could be that the digital and software designs give a lot more flexibility to the engineers. They have more options in terms of profiles, projects and companies.

Due to the scarcity, the salary demand of these engineers is a lot higher. When they come out of the engineering school, they make \$10,000, and within a few years they demand \$100,000, which is a lot more than in Silicon Valley.

In India, the focus has been more on software. Even in semiconductor companies that have R&D centres in India, such TI and STMicroelectronics, more of digital and software designing is done. The fact that the government also does not have a department of electronics but department of IT and communications shows the change in focus. In a way this is good as companies are making money, but from analogue perspective the growth gets stagnated.

In analogue design, engineers need to work together. As it cannot be in the form of modules, a lot of interaction is required. Also, as the work culture of changing jobs and moving up the ladder prevails in India, developing expertise in analogue design could be challenging. ●